SERIES REACTIVE CIRCUITS

AC CIRCUITS

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GRADING SYSTEM

FINAL GRADE

30%	30%	**40%**
PRELIM	MIDTERM	ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Phasor Diagram

Series R-L Circuit

Series R-C Circuit

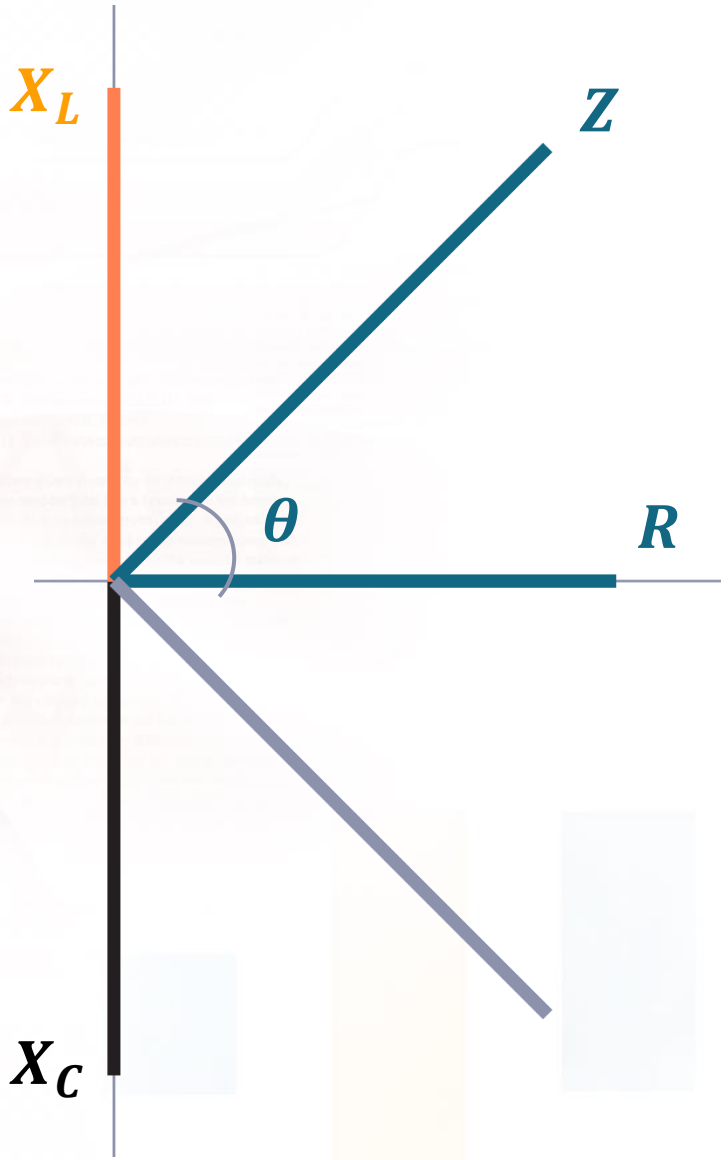
Series R-L-C Circuit

Power Factor

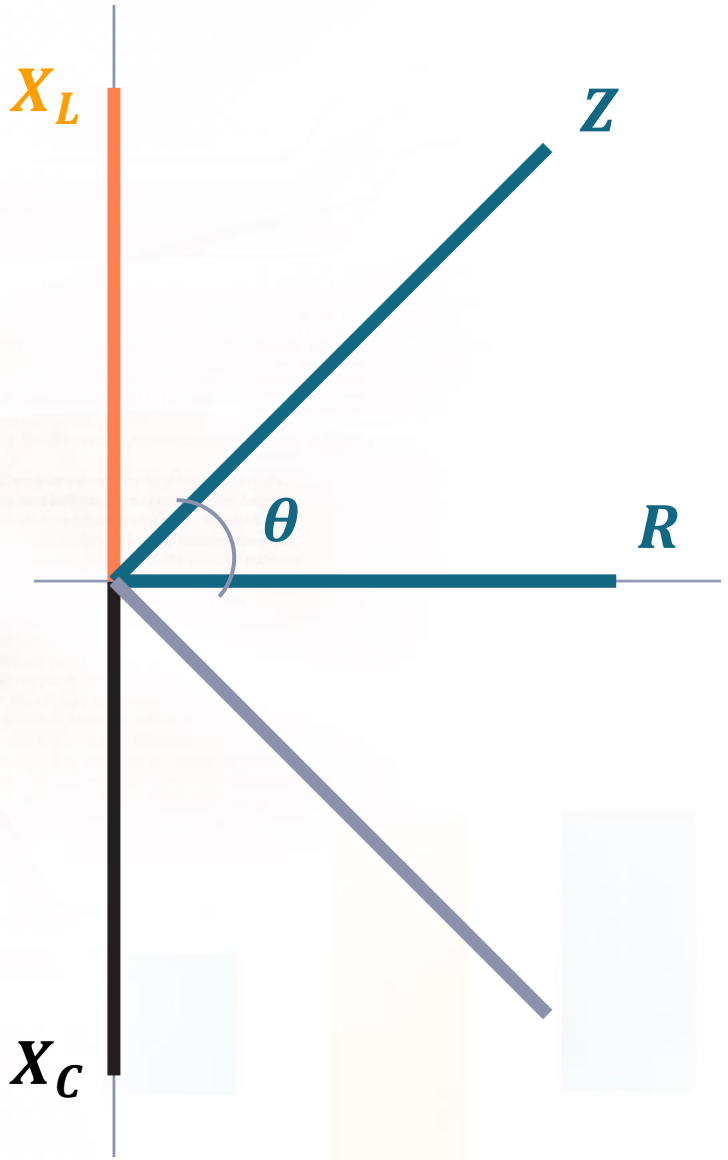
PHASOR DIAGRAM

PHASOR DIAGRAM

Graphical representation of **magnitude** and **phase** relationship between sinusoidal quantities

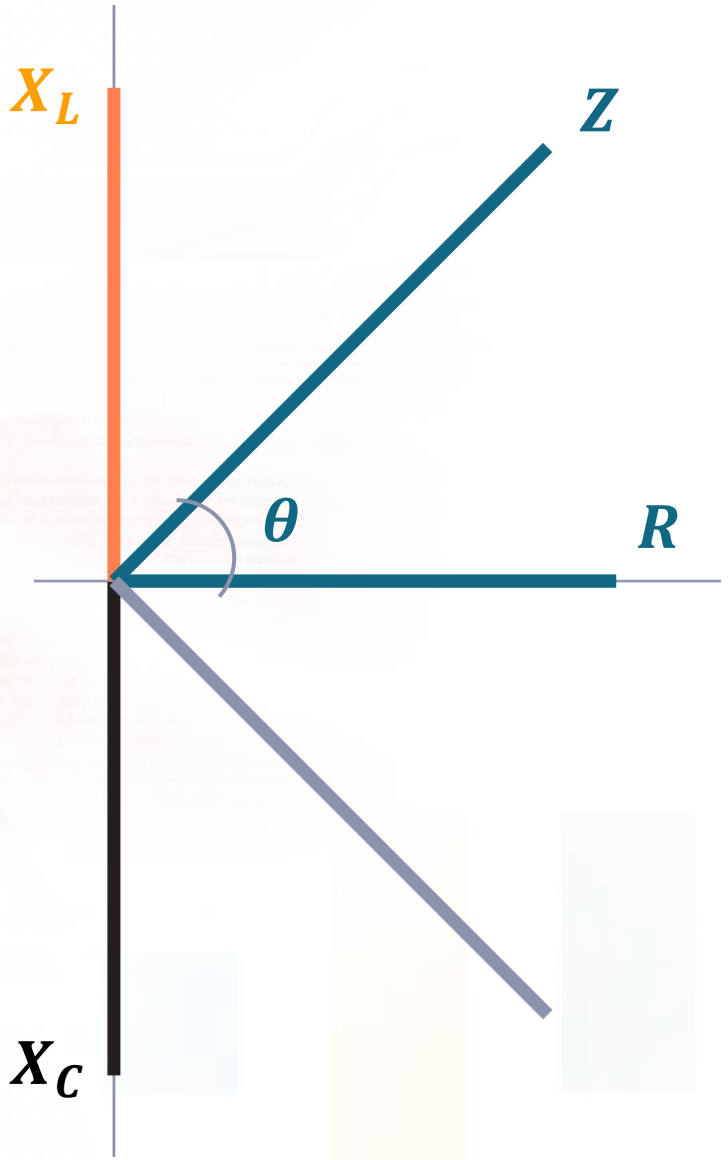


Z, represents the **total opposition** offered by circuit elements (including resistance and reactance) to the flow of alternating current (**AC**)



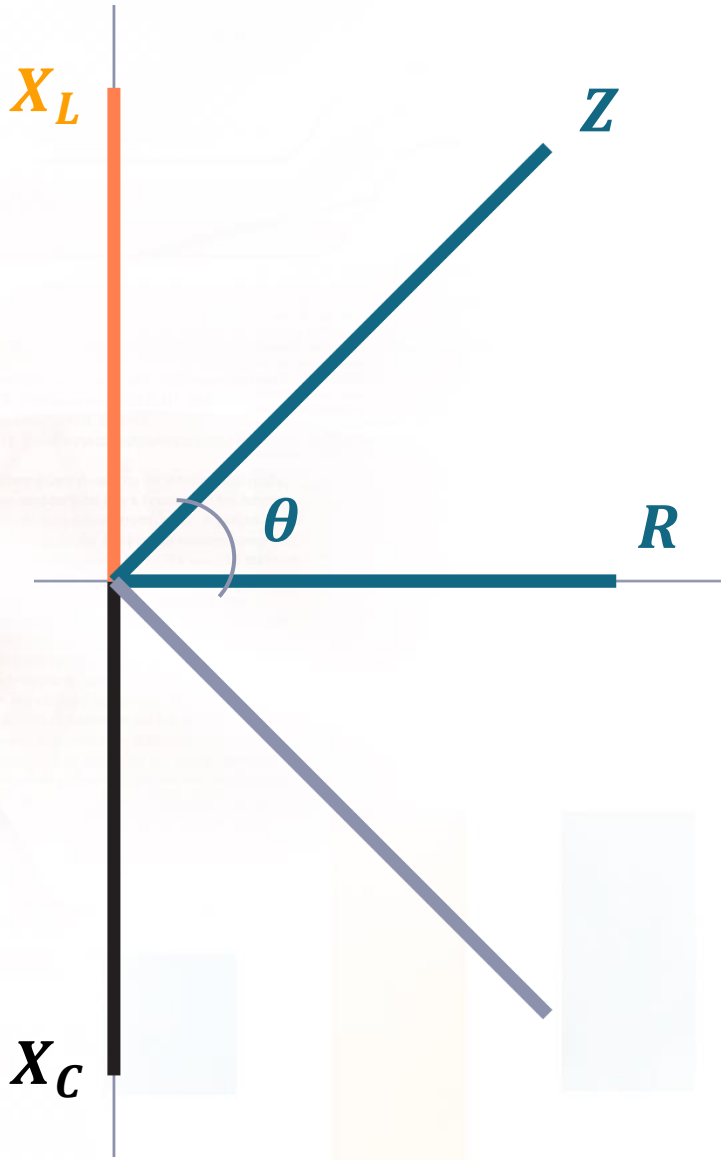
INDUCTIVE REACTANCE

X_L, the **opposition** offered by the **inductor** to the flow of alternating current (**AC**)



CAPACITIVE REACTANCE

X_c, the **opposition** offered by the **capacitor** to the flow of alternating current (**AC**)



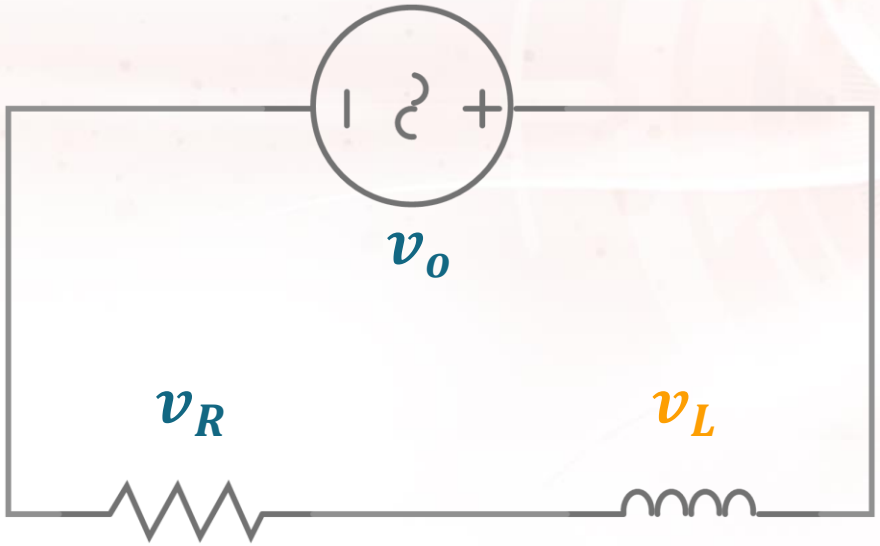
SERIES R-L CIRCUIT

TOTAL INDUCTANCE

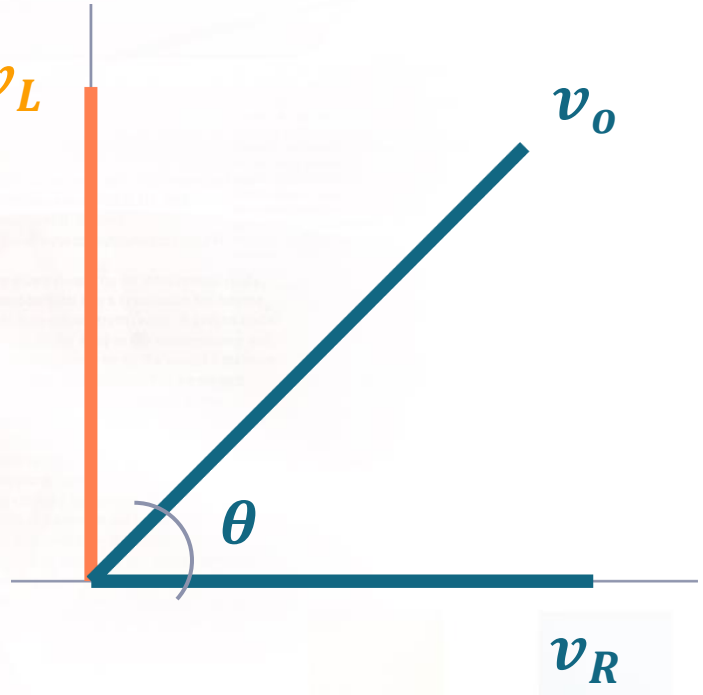
Series,

$$L_o = L_1 + L_2 + L_3 + \cdots L_n$$

$$v_o = v_R + jv_L$$



Circuit Diagram



Voltage Phasor Diagram

INDUCTIVE REACTANCE

$$X_L = 2\pi f L$$

The **opposition** offered by the **inductor** to the flow of alternating current (AC)

Directly proportional to the frequency

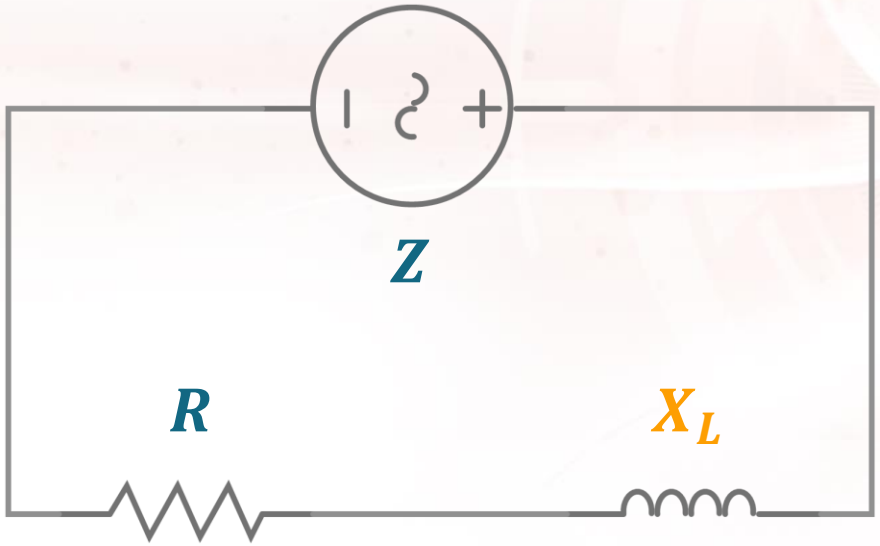
where:

$X_L = inductive\ reactance, \Omega$

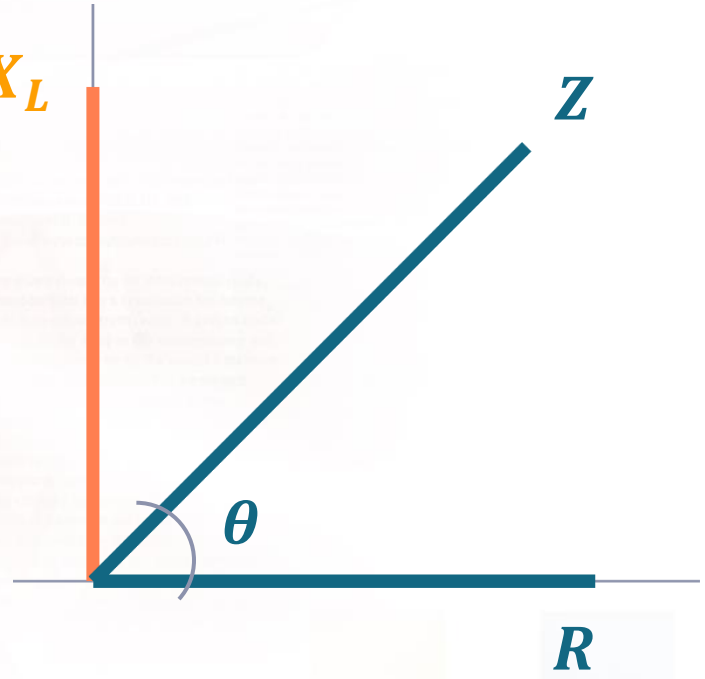
$f = frequency, Hz$

$L = inductance, H$

$$Z = R + jX_L$$



Circuit Diagram



Impedance Phasor Diagram

A **240-volt 60-cycle** source is connected to a coil of wire that has a resistance of **7.5 ohms** and an inductance of **0.0477 henry**.

Calculate the following:

a) impedance, Z

b) current, i_o

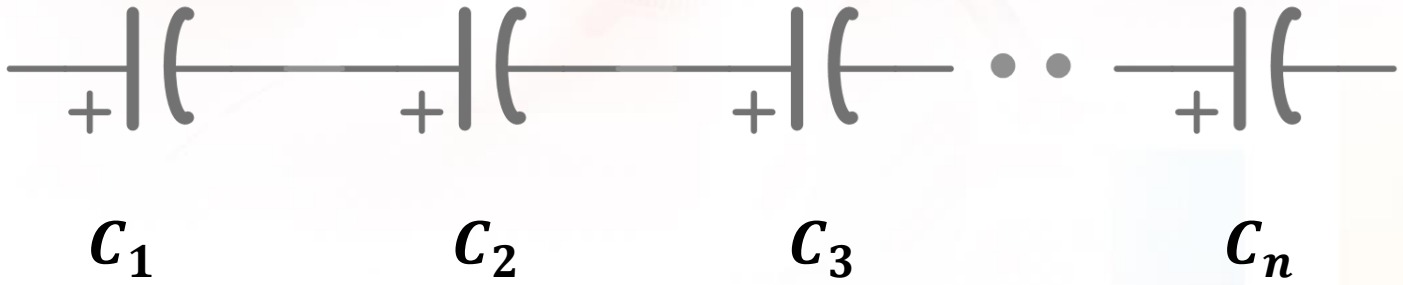
c) voltage across the resistor, v_R

d) voltage across the inductor, v_L

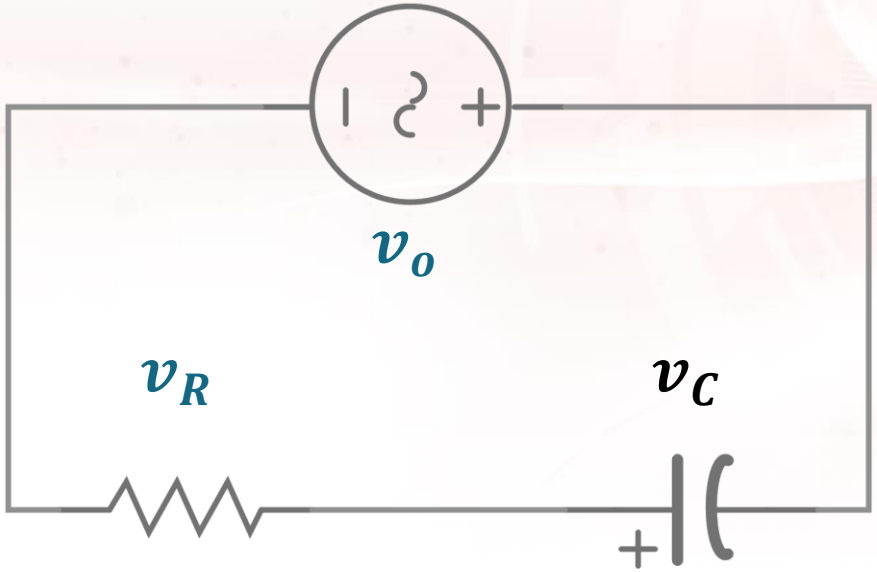
SERIES R-C CIRCUIT

Series,

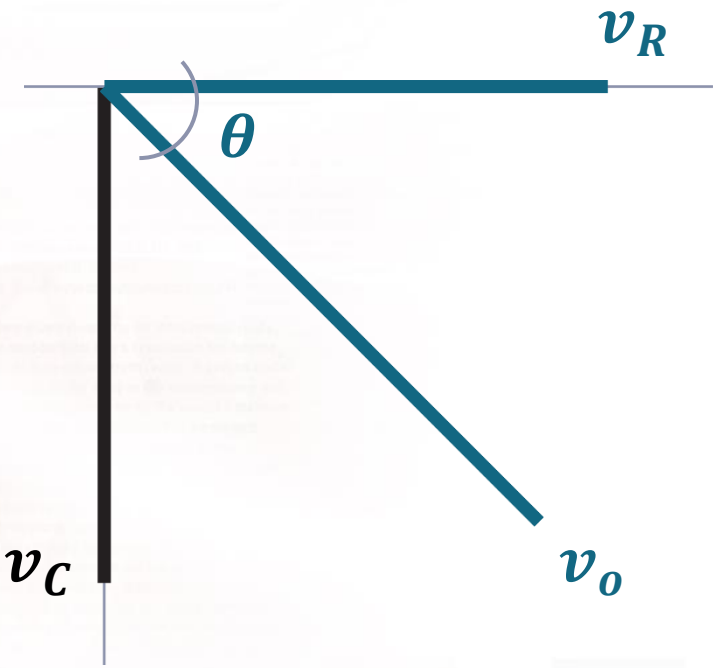
$$\frac{1}{C_o} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \frac{1}{C_n}$$



$$v_o = v_R + jv_C$$



Circuit Diagram



Voltage Phasor Diagram

CAPACITIVE REACTANCE

$$X_C = \frac{1}{2\pi f C}$$

The **opposition** offered by the **capacitor** to the flow of alternating current (AC)

Inversely proportional to the frequency

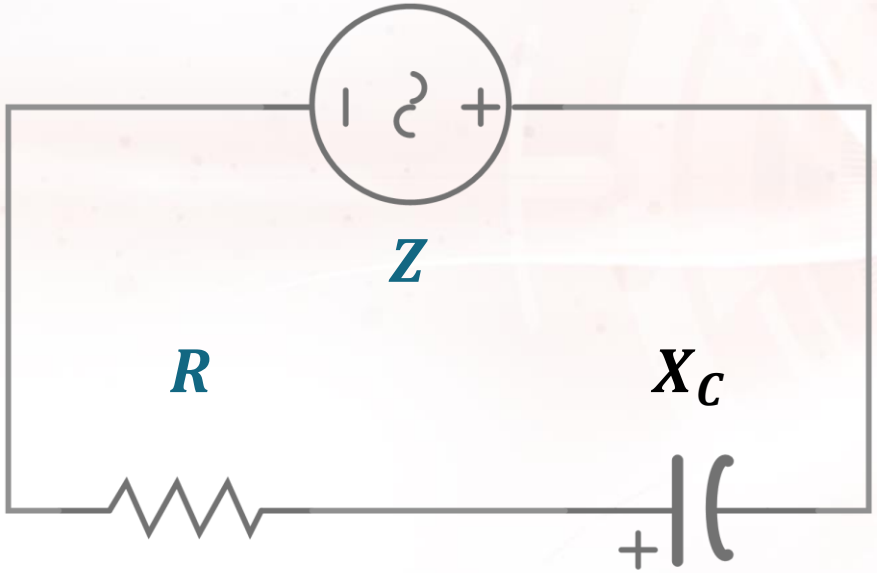
where:

X_C = capacitive reactance, Ω

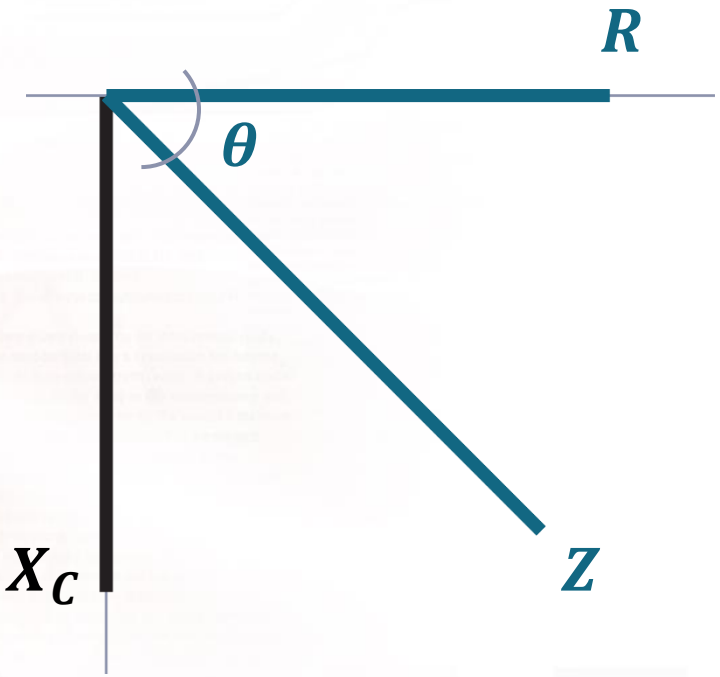
$f = frequency, Hz$

$C = capacitance, F$

$$Z = R + jX_C$$



Circuit Diagram



Impedance Phasor Diagram

A **125-volt 25-cycle** source is connected to a series circuit consisting of a **30-ohm 159-μf** capacitor.

Calculate the following:

a) impedance, Z

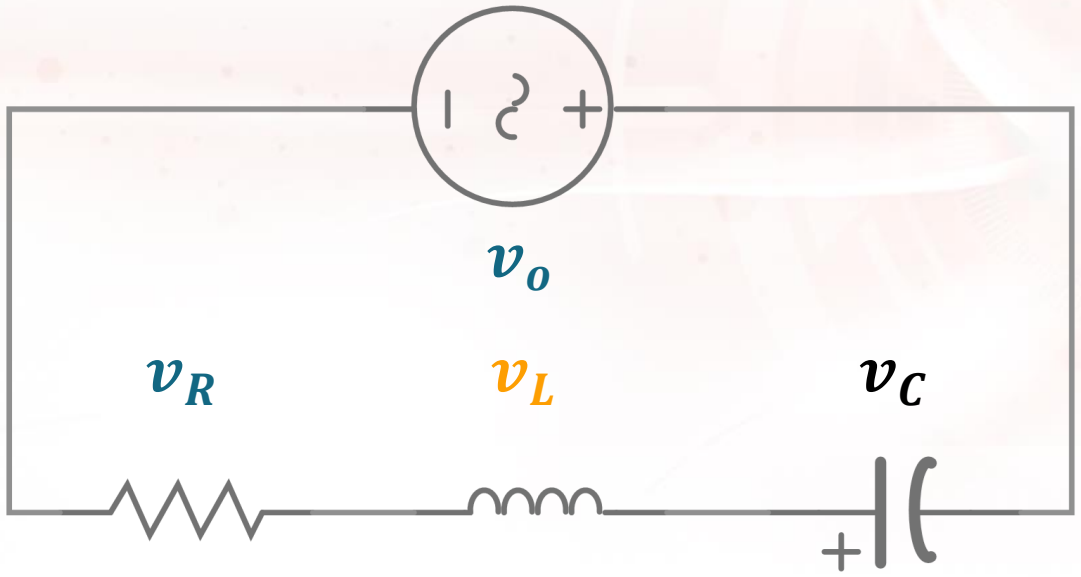
b) current, i_o

c) voltage across the resistor, v_R

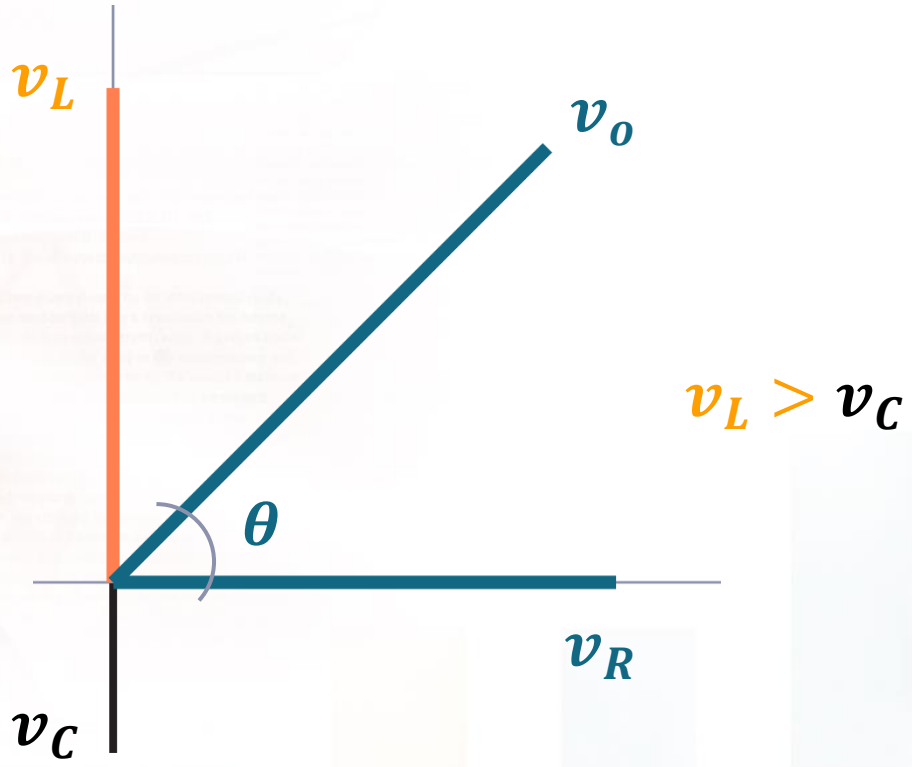
d) voltage across the capacitor, v_C

SERIES R-L-C CIRCUIT

$$v_o = v_R + j(v_L - v_C)$$

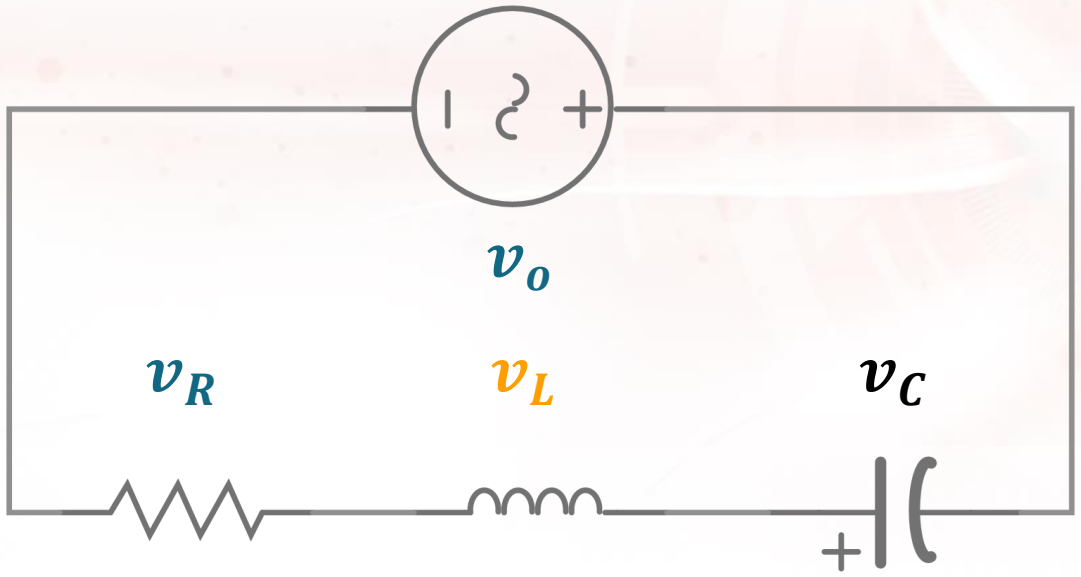


Circuit Diagram

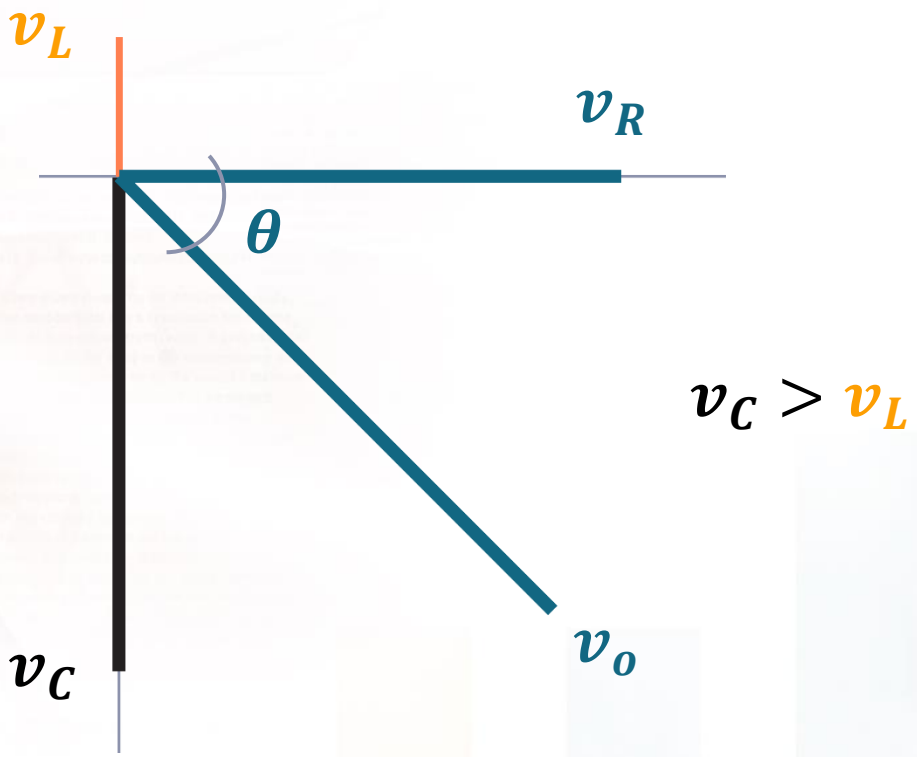


Voltage Phasor Diagram

$$v_o = v_R + j(v_L - v_C)$$



Circuit Diagram



Voltage Phasor Diagram

A series circuit consisting an **80-ohm** resistor, a **0.3-henry** inductor, and a **50-μf** capacitor is connected to a **120-volt, 60-cycle** source.

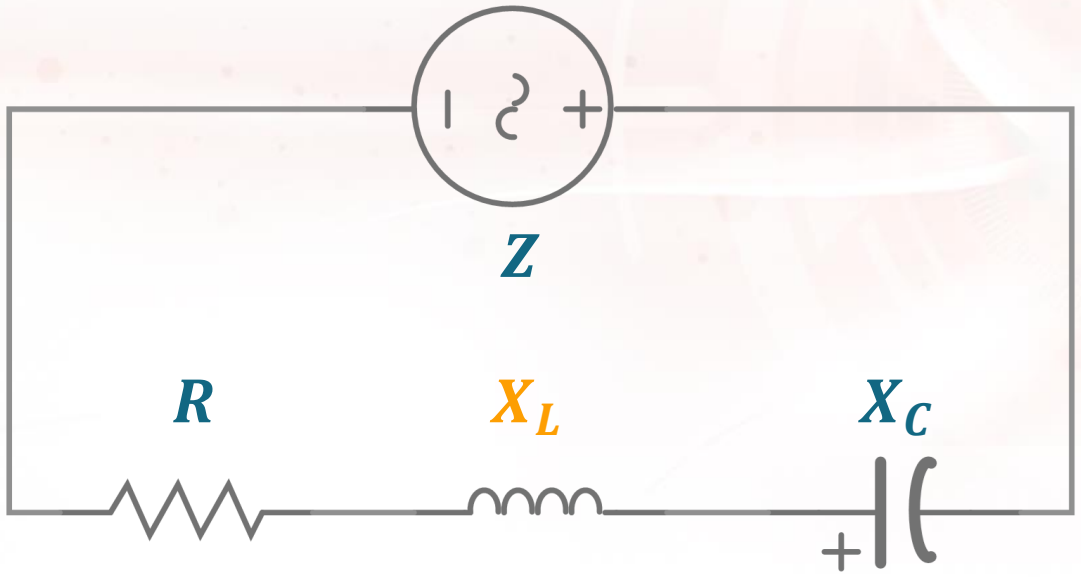
Calculate the following:

a) equivalent impedance, Z

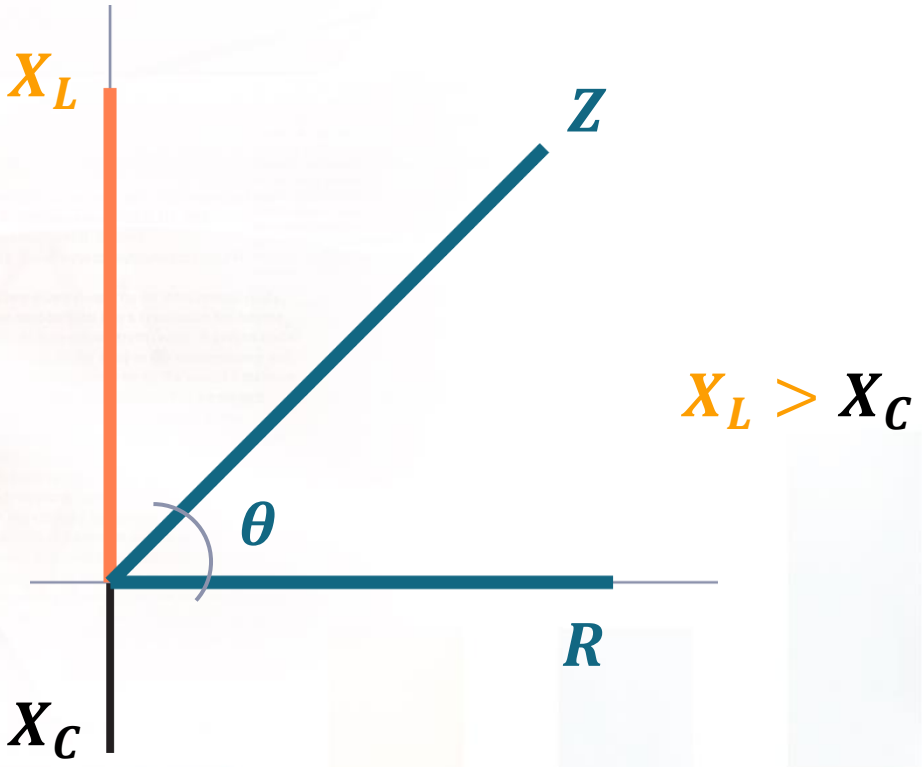
b) Current, i_o

c) voltage drops, v_R, v_c, v_L

$$Z = R + j(X_L - X_C)$$

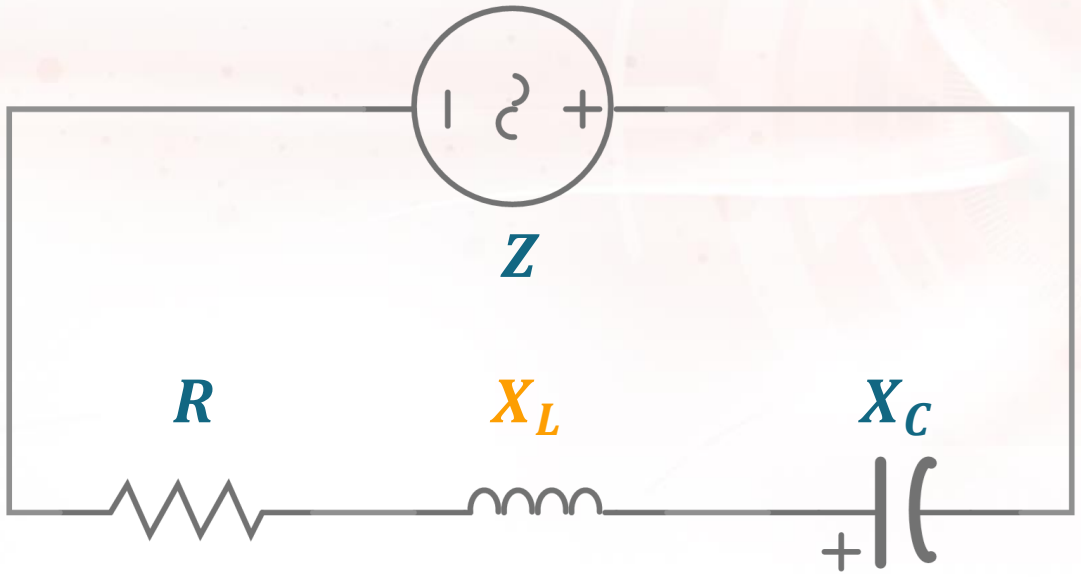


Circuit Diagram

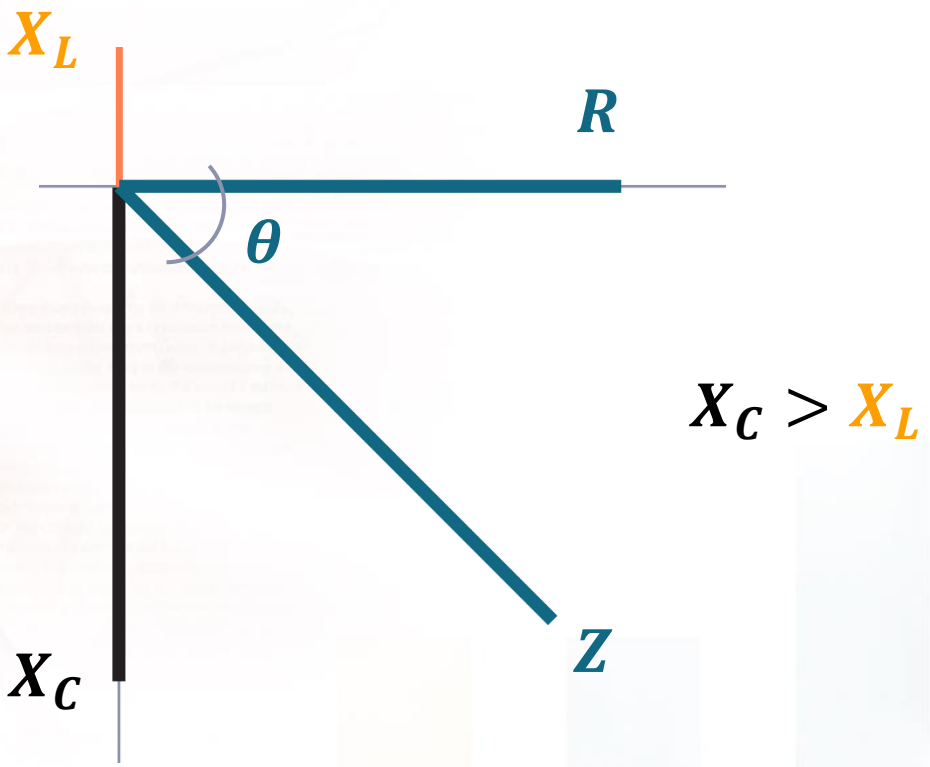


Impedance Phasor Diagram

$$Z = R + j(X_L - X_C)$$



Circuit Diagram



Impedance Phasor Diagram

EXERCISE

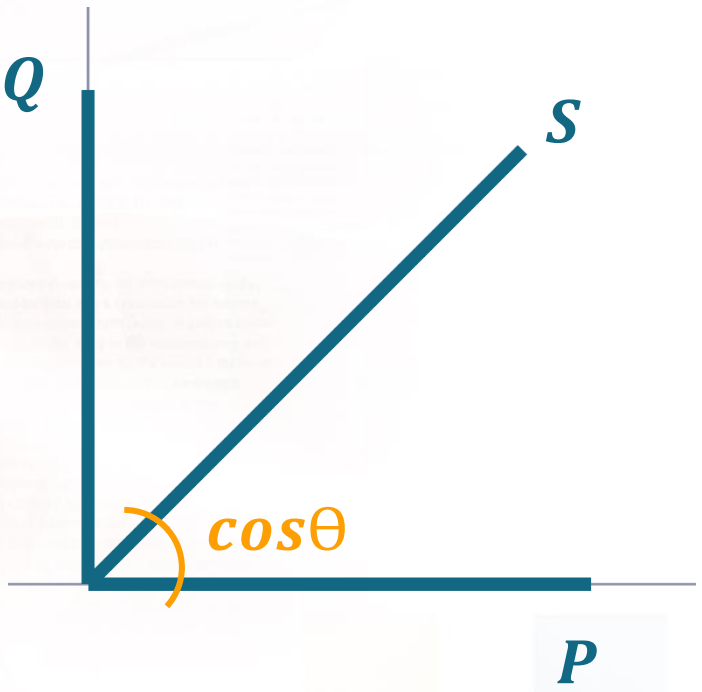
A series circuit consisting of a **30-μF** capacitor, and a **0.155-henry** inductor is connected to a **120-volt 60-cycle** source. Calculate the circuit **current** and indicate whether it **lags** or **leads** the voltage.

If a variable **inductor** is substituted for the one in the previous problem, what should be its value if an equal **current is to lag** behind the **voltage**? Assume all other conditions to remain unchanged.

POWER FACTOR

The ratio of true power to apparent power

$$\cos\theta = \frac{P}{S}$$



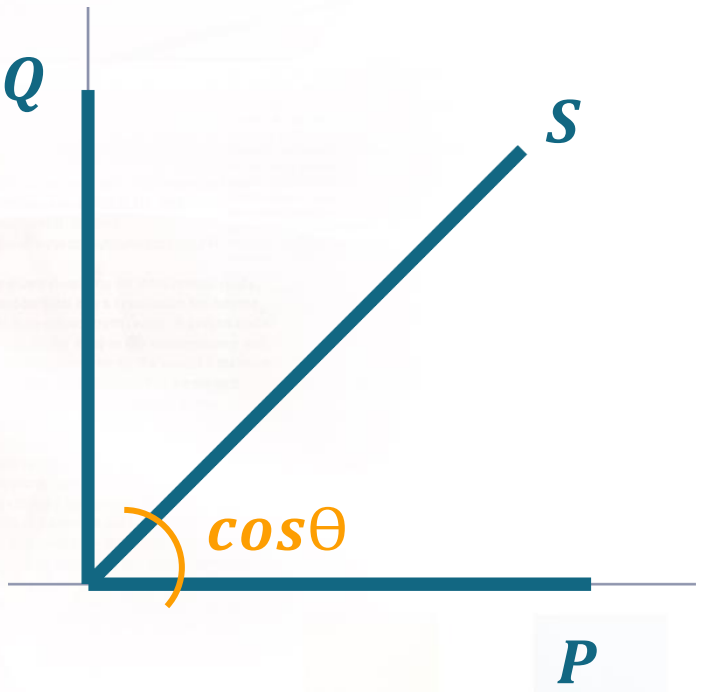
True Power, P

$$P = i^2 R$$

unit: watt (W)

The ratio of true power to apparent power

$$\cos\theta = \frac{P}{S}$$



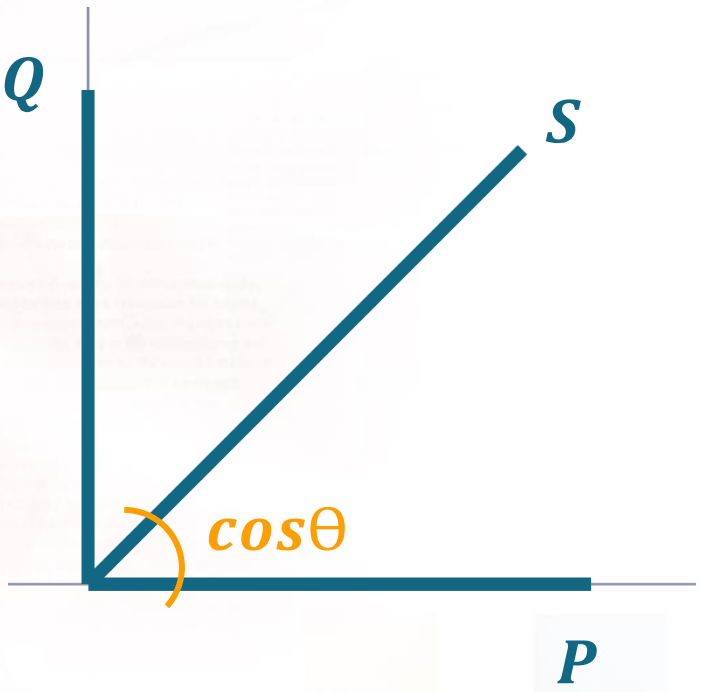
Reactive Power, Q

$$Q = i^2 X$$

unit: Volt-Amps-Reactive (VAR)

The ratio of true power to apparent power

$$\cos\theta = \frac{P}{S}$$



Apparent Power, S

$$S = i^2 Z$$

unit: Volt-Amps (VA)

EXERCISE

A series circuit consisting an **80-ohm** resistor, a **0.3-henry** inductor, and a **50-μf** capacitor is connected to a **120-volt, 60-cycle** source.

Calculate the **power** and **power factor**.

LABORATORY